

## Thermal: VRU NVL72 to Boost Cold Plate &amp; QD Value

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## What's New?

Structurally, the rising AI server TDP and power consumption are driving greater adoption of advanced cooling solutions and higher cooling content per system. For 2027, we expect the trend to continue into Rubin Ultra platform, despite HBM de-spec, driven by the likely double-sided of cold plate per GPU and enhanced material of copper diamond. We also expect QD value to increase.

## Comments

**Rising AI server power consumption drives cooling content:** GPU TDP has surged from Ampere/Hopper's 400W–700W to Blackwell's 1,000–1200W, with Rubin expected to reach up to 2,300W; AI server racks have scaled from 10kW (H200) to 36kW (GH200) in earlier GPU deployments to 130–140kW (GB300) in current configurations, and expected to head towards 600kW to 1MW in future generations. This shift makes AI server platforms increasingly depend on advanced cooling solutions, driving higher cooling content for a system.

**Cold plate content per rack to be driven by double-sided and copper diamond:** Due to higher TDP (we expect 2,600W to be mainstream for Rubin Ultra), VR200 NVL72 includes both large and small form-factor cold plates within the compute tray and upgrades to a microchannel cold plate (MCCP) architecture; hence, we expect VR200 NVL72 to utilize 5 cold plates in each compute tray (2 large plates and 3 small plates), with switch tray remaining the same, leading to the content per rack of ~\$34,200, slightly increase from that of GB300 NVL72. For VR300 (Rubin Ultra), we expect the content value to further increase by 58%, vs. that of VR200 NVL72, driven by: 1) double-sized cold plate for GPU, 2) copper diamond powder/material for GPU's cold plate; 3) doubled units of switch ASICs (for one NVL72).

**QD content per rack to rise into Rubin platform:** Given Vera Rubin shifts to all liquid-cooled architecture, VR200 NVL72 expands the number of liquid-cooled components, driving more QD use per system. We estimate VR200 NVL72 to utilize 22 QDs in each compute tray and 6 QDs in each switch tray, resulting in content per rack of ~\$10,760, 77% increase from GB300 NVL72. For VR300, we expect the content value to further increase by 103%, vs. that of VR200 NVL72, thanks to upgrade to ZQD and doubled usage per switch tray (due to 2 boards per tray).

## Key beneficiaries

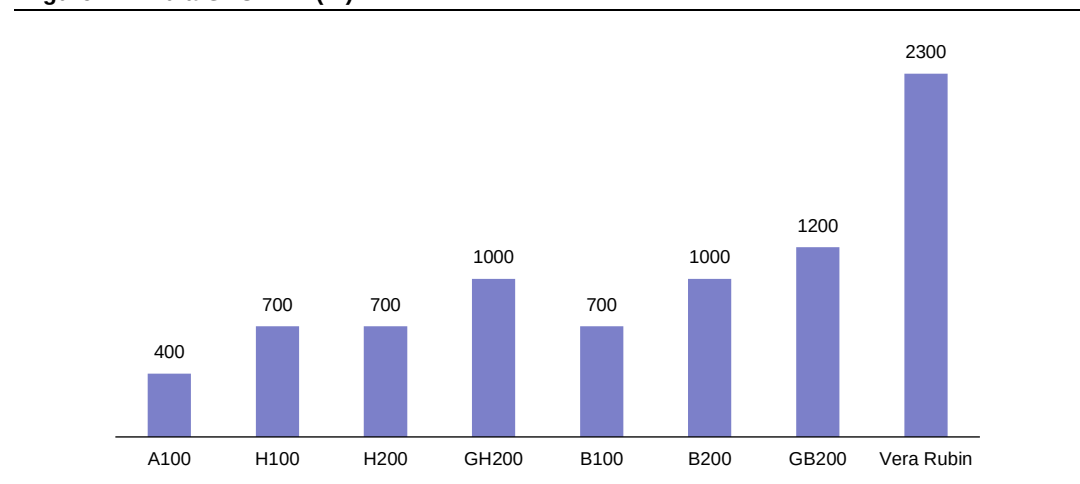
- **Cold plate:** AVC (3017 TT)
- **QD:** Fositek (6805 TT) with 50%+ share, LOTES (3533 TT), FIT (06088 HK Buy)
- **ODMs/OEMs:** Hon Hai (2317 TT Buy), Supermicro (SMCI US Buy)

**Risks:** 1) AI demand slowdown, 2) infrastructure deployment, 3) supply constraints, 4) competition.

## Rising TDP and power consumption for AI server

AI server power content is rising structurally, driven by increasing system/rack level power demands along with expanding per-chip power envelopes. While conventional host server CPU TDPs typically peak at 300–500W, advanced AI accelerators have scaled aggressively across generations. Take Nvidia's GPU superchips for instance, Single-GPU TDP has surged from legacy Ampere/Hopper's 400W–700W to Blackwell's 1,000-1200W, with upcoming Rubin expected to reach up to 2,300W. However, due to HBM de-spec, we believe the Rubin Ultra series' TDP could be in the range of 1,800 to 2,600. Yet, we believe next-gen platforms of Feynman will increase TDP even further, sustaining the demand for higher-density power delivery.

**Figure 1: Nvidia GPU TDP (W)**



Sources: Company data, GF Securities (Hong Kong) Brokerage

At system level, rising accelerator TDP directly translates into a step-up in rack-level power consumption. AI server racks have scaled from 10kW (H200) to 36kW (GH200) in earlier GPU deployments to 130–140kW (GB300) in current configurations, and expected to head towards 600kW to 1MW in future generations. This shift makes AI server platforms increasingly depend on advanced cooling solutions, driving higher cooling content for a system.

**Figure 2: Nvidia rack solutions roadmap**

|                    | 2022-2023   | 2024        | 2025            | 2026                  | 2027E        | 2028E      |
|--------------------|-------------|-------------|-----------------|-----------------------|--------------|------------|
| Platform           | GH100/200   | GB200       | GB300           | VR200                 | VR300        |            |
| GPU                | Hopper      | Blackwell   | Blackwell Ultra | Vera Rubin            | Rubin Ultra  | Feynman    |
| Rack Configuration | NVL 36      | NVL 72      | NVL 72          | NVL 144               | NVL 576      |            |
| AC/DC              | 400V/50V    | 400V/50V    | 400V/50V        | 400V/50V              | 400V/800V    | 400V/800V  |
| Power Architecture | Power Shelf | Power Shelf | Power Shelf     | Power Shelf / Sidecar | HVDC Sidecar | HVDC / SST |
| Power Consumption  | 36kW        | 120-130 kW  | 130-140kW       | 220-230kW             | TBD          | TBD        |

Sources: Tech Insights, GF Securities (Hong Kong) Brokerage

### Cooling content per rack to rise into Rubin platform

As said, cooling architecture has transitioned from a routine supporting function to a system-level performance bottleneck, marking a long-term structural upgrade in AI server cooling design and continuous increase of content value, where Blackwell supports both liquid and air-cooled configurations, while Vera Rubin shifts to an all liquid-cooled architecture, including GPU/CPU to components such as SOC/AMM and NIC/ConnectX.

For cold plate, VR200 NVL72 includes both large and small form-factor cold plates within the compute tray and upgrades to a microchannel cold plate (MCCP) architecture; hence, we expect VR200 NVL72 to utilize 5 cold plates in each compute tray (2 large plates and 3 small plates), vs. 6 small plates for GB300 NVL72, while blended ASP rise to ~\$292 from ~\$240, with switch tray remaining the same, leading to the content per rack of ~\$34,200, slightly increase from that of GB300 NVL 72. For VR300 (Rubin Ultra), we expect the content value to be further increase by 58% with more use of small plates, vs. that of VR200 NVL72, driven by: 1) double-sized cold plate for GPU, 2) copper diamond powder/material for GPU's cold plate; 3) doubled units of switch ASICs (for one NVL72).

For QD (Quick Disconnect), Given Vera Rubin shifts to all liquid-cooled architecture, VR200 NVL 72 materially expands the number of liquid-cooled components, driving more QD use per system. We estimate VR200 NVL72 to utilize 22 QDs in each compute tray and 6 QDs in each switch tray, resulting in content per rack of ~\$10,760, 77% increase from GB300 NVL72. For VR300, we expect the content value to increase by 103% thanks to upgrade to ZQD and doubled usage per switch tray (due to 2 boards per tray).

Figure 3: Cold plate content forecast

|                       | GB300 NVL72 | VR200 NVL72 |
|-----------------------|-------------|-------------|
| TDP (kW)              | 120         | 230         |
| Units                 | 135         | 117         |
| Blended ASP (\$)      | 240         | 292         |
| Content per Rack (\$) | 32,400      | 34,200      |
| Increment             |             | 6%          |

Sources: GF Securities (Hong Kong) Brokerage

Figure 4: QD content forecast

|                       | GB300 NVL72 | VR200 NVL72 |
|-----------------------|-------------|-------------|
| TDP (kW)              | 120         | 230         |
| Units                 | 304         | 538         |
| Blended ASP (\$)      | 20          | 20          |
| Content per Rack (\$) | 6,080       | 10,760      |
| Increment             |             | 77%         |

Sources: GF Securities (Hong Kong) Brokerage

**Figure 5: Cold plate and QD content forecast for VR300 NVL 72**

|                             | Cold Plate | QD     |
|-----------------------------|------------|--------|
| Units                       | 216        | 808    |
| Blended ASP (\$)            | 250        | 27     |
| Content per Rack (\$)       | 54,000     | 21,880 |
| Increment from VR200 NVL 72 | 58%        | 103%   |

Sources: GF Securities (Hong Kong) Brokerage

## Risks

1) AI demand slowdown, 2) infrastructure deployment, 3) supply constraints, 4) competition.

## Rating definitions Benchmark: Hang Seng Index (Hong Kong)

|                        |                     |                                                                 |
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|                       |                 |                                                                  |
|-----------------------|-----------------|------------------------------------------------------------------|
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